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University of Benin, Benin City, Faculty of Engineering,
Electrical/Electronic Engineering Department
First Semester B.Eng. (Honours) Degree Examination 2024/2025 Academic Session

Course Code: EEE571 *Course Title: Electronic Circuit II*
Date of Examination: 30/05/2025 *Exam Duration: 3hrs*
Instruction(s): Answer five questions with three in section A and two in section B.
Examiners: Engr. Dr. O.S Omorogiuwa and Engr. Dr. E. Osa. Note don't write on your question paper.

Section A

Question 1

(a) When the ac input voltage of a certain power supply changes, the input to the voltage regulator decreases by 3 V as a result, and the output of the regulator decreases by 0.5 V. The nominal output is 15 V. Determine the line regulation in %/V. 1mark 1

(b) A certain voltage regulator has a 12 V output when there is no load, when there is a full-load, output voltage is 11.9 V. Express the voltage regulation as a percentage change from no-load to full-load. 1mark 1

(c) Using diagram, explain Short-Circuit Protection in regulated power supply. 4marks 3

(d) With simple diagram, explain the operation of negative resistance oscillator using tunnel diode? 4marks

(e) Using simple diagram explain the term adjustable negative linear regulator 4marks 3

Question 2

(a) In a centre-tap full-wave rectifier, the load resistance $R_L = 1k\Omega$ each diode has a forward-bias dynamic resistance r_d of 10Ω. The voltage across half the secondary winding is (220sin 314t)V. Find

(i) the peak value of current,

(ii) the average value of current,

(iii) the r.m.s value of current,

(iv) the ripple factor, 2

(v) the rectification efficiency. 5.5marks

(b) A BJT supplies 0.85 W to a 4K Ω load. The zero signal dc collector current is 31 mA. Determine the percent second harmonic distortion. 2.5marks 2

(c) With simple diagram, explain push-pull amplifier principle 4marks 3

(d) State the two Barkhausen criterion for an oscillator. 2marks 2

Question 3

(a) Design a shunt regulator using zener diode to provide 5V dc voltage for a variable load of 100Ω to 500 Ω. The minimum zener current for its proper operation is 5 mA. The unregulated input voltage = 6 V. 4.5marks

(b) With simple diagram, explain Clapp oscillator and derive the expression for frequency of oscillation. 4.5marks

(c) What is Blocking Oscillator? 2marks

(d) A class A power amplifier with a direct coupled load has a collector efficiency of 30% and delivers a power of 10W. Find

(i) the dc power input

(ii) the power dissipation of full output and

(iii) the desirable power dissipation rating of the BJT 3marks

Question 4

(a) Design a series pass regulator to obtain 10 V dc output with maximum load current of 100 mA. Consider the nominal input voltage to be 15 V. 3marks 1

(b) With simple circuit diagram, explain emitter coupled astable Multivibrator. 3marks 3

(c) An Astable Multivibrator circuit is constructed using two timing capacitors of equal value of 3.3uF and two base resistors of value 10kΩ. Calculate the minimum and maximum frequencies of oscillation if a 100kΩ dual-gang potentiometer is connected in series with the two resistors. 2marks

(d) What are Time Base generators? 1marks 3

(e) For the transformer coupled class A power amplifier circuit derive the expression for its efficiency. 3marks

(f) An Astable Multivibrators circuit is required to produce a series of pulses at a frequency of 500Hz with a mark-to-space ratio of 1:5. If $R_2 = R_3 = 100k\Omega$, calculate the values of the capacitors, C_1 and C_2 required. 2marks

Section B

Question 5

(a.) Describe in sufficient details the drawbacks of the differential amplifier? 4marks

(b.) With the aid of appropriate circuit diagrams and detailed explanations, describe how the limitations of the basic differential amplifier can be overcome. 10marks

Question 6

(a.) Explain using appropriate circuit diagram the Op-Amp as a five terminal device. 4marks

(b.) Using relevant circuit diagrams and equations realize the output voltage expression for the instrumentation amplifier circuit. 10marks

Question 7

(a.) With the aid of appropriate circuit diagram and equations, design the Op-amp circuit application that gives the following output voltage expression: $V_o = V_{in}(1 + \frac{R_f}{R_1})$. 7marks

(b.) (i.) Design an inverting comparator circuit with V_{ref} applied at the non-inverting input.

(ii.) Show the input and output waveforms when $V_{ref} > 0$. 7 marks